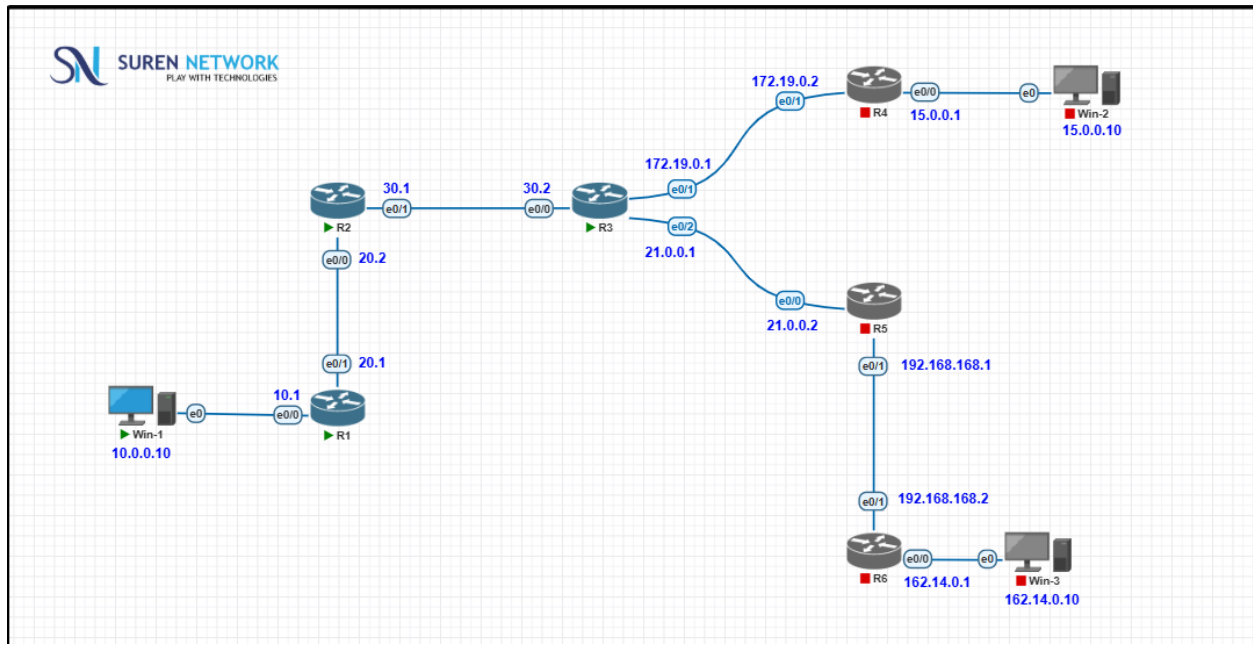
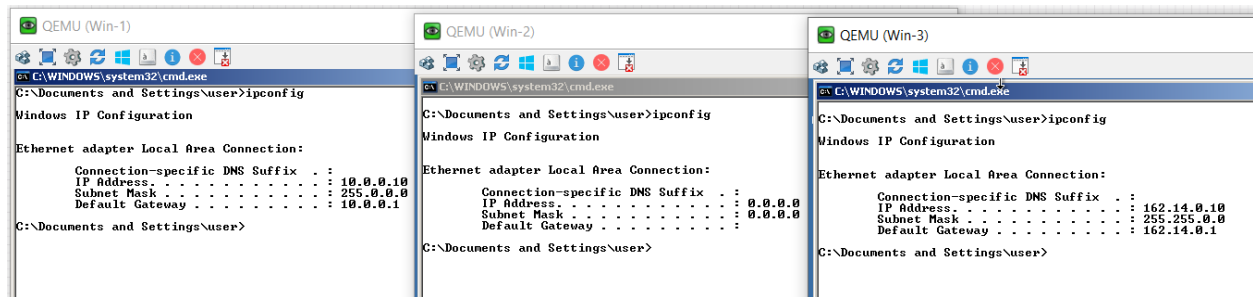


Static Route Project



Ip configuration In PCs



Configure static Route in all cisco routers

```
R1
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
S    15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    21.0.0.0/8 [1/0] via 20.0.0.2
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    162.14.0.0/16 [1/0] via 20.0.0.2
S    172.19.0.0/16 [1/0] via 20.0.0.2
S    192.168.168.0/24 [1/0] via 20.0.0.2
```

 R2

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.19.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
```

R2(config)#

 R3

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/2
L     21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/0
L     30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.19.0.0/16 is directly connected, Ethernet0/1
L     172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```

R3(config)#

 R4

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     15.0.0.0/8 is directly connected, Ethernet0/0
L     15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.19.0.0/16 is directly connected, Ethernet0/1
L     172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

R4(config)#

```

R5
S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/0
L    21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.1/32 is directly connected, Ethernet0/1
R5(config)#

```

```

R6
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    162.14.0.0/16 is directly connected, Ethernet0/0
L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.2/32 is directly connected, Ethernet0/1
R6(config)#

```

Connectivity Check

Ping Win-1 to Win-2 -Success

QEMU (Win-1)

```

C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 50.0.0.10
'ping' is not recognized as an internal or external command,
operable program or batch file.

C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

```

QEMU (Win-2)

```

C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 15.0.0.10
    Subnet Mask . . . . . : 255.0.0.0

```

Ping Win-2 to Win-3 -Success

QEMU (Win-2)	QEMU (Win-3)
<pre>C:\WINDOWS\system32\cmd.exe C:\Documents and Settings\user>ping 162.14.0.10 Pinging 162.14.0.10 with 32 bytes of data: Reply from 162.14.0.10: bytes=32 time=2ms TTL=124 Reply from 162.14.0.10: bytes=32 time=2ms TTL=124 Reply from 162.14.0.10: bytes=32 time=2ms TTL=124 Reply from 162.14.0.10: bytes=32 time=2ms TTL=124 Ping statistics for 162.14.0.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 2ms, Average = 2ms C:\Documents and Settings\user></pre>	<pre>C:\WINDOWS\system32\cmd.exe Windows IP Configuration Ethernet adapter Local Area Connection: Connection-specific DNS Suffix . : IP Address. : 162.14.0.10 Subnet Mask : 255.255.0.0 Default Gateway : 162.14.0.1 C:\Documents and Settings\user>ping 15.0.0.10 Pinging 15.0.0.10 with 32 bytes of data: Reply from 15.0.0.10: bytes=32 time=4ms TTL=124 Reply from 15.0.0.10: bytes=32 time=2ms TTL=124 Reply from 15.0.0.10: bytes=32 time=3ms TTL=124 Reply from 15.0.0.10: bytes=32 time=2ms TTL=124 Ping statistics for 15.0.0.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 4ms, Average = 2ms C:\Documents and Settings\user></pre>

Ping Win-1 to Win-3 -Success

QEMU (Win-1)	QEMU (Win-3)
<pre>C:\WINDOWS\system32\cmd.exe C:\Documents and Settings\user>ping 162.14.0.10 Pinging 162.14.0.10 with 32 bytes of data: Reply from 162.14.0.10: bytes=32 time=2ms TTL=123 Reply from 162.14.0.10: bytes=32 time=4ms TTL=123 Reply from 162.14.0.10: bytes=32 time=2ms TTL=123 Reply from 162.14.0.10: bytes=32 time=2ms TTL=123 Ping statistics for 162.14.0.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 4ms, Average = 2ms C:\Documents and Settings\user></pre>	<pre>C:\WINDOWS\system32\cmd.exe C:\Documents and Settings\user>ping 10.0.0.10 Pinging 10.0.0.10 with 32 bytes of data: Reply from 10.0.0.10: bytes=32 time=2ms TTL=123 Reply from 10.0.0.10: bytes=32 time=3ms TTL=123 Reply from 10.0.0.10: bytes=32 time=3ms TTL=123 Reply from 10.0.0.10: bytes=32 time=1ms TTL=123 Ping statistics for 10.0.0.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 3ms, Average = 2ms C:\Documents and Settings\user></pre>